



Vasavi College of Engineering

Autonomous | NAAC 'A++' grade | NIRF-2024 (Band 201-300)

All U.G. Programmes are accredited by NBA

Affiliated to Osmania University | Approved by AICTE.

CAPSTONE PROJECT 2025- MECHANICAL ENGINEERING DEPT.



MAKE IN INDIA

An Initiative by The Government of India

RENUKA
PRECISION SYSTEMS

LOCAL INDUSTRY COLLABORATOR

BATCH 12- KANAMALURI SAI VENKAT TUSHAR
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PROJECT TITLE - ERP-Integrated Manufacturing & Supply Chain Control Tower

INTRODUCTION

INDUSTRY COLLABORATOR BACKGROUND-Situated in Hyderabad , India ; Renuka Precision systems (RPS) is a high-precision machining company serving industrial equipment, automotive, and heavy machinery sectors worldwide. The organization operates CNC turning, milling, grinding, and inspection processes while managing inventory, procurement, supplier performance, and production planning. Increasing operational complexity highlighted the need for an integrated decision-support platform capable of improving visibility and supporting data-driven decision-making across the manufacturing value chain.

PROBLEM STATEMENT:

- Supplier lead-time variability and sourcing risks
- Excess Inventory and Working Capital Utilization
- Procurement inefficiencies and limited spend visibility
- Production scheduling and capacity constraints
- Disconnected operational information systems

Objective: Create a centralized platform integrating inventory, procurement, supplier, and production data into a unified decision-support framework.

CONTROL TOWER MODULES

**DATA
COLLECTION**



**INVENTORY
OPTIMIZATION**



**SUPPLIER
ANALYTICS**



**PROCUREMENT
ANALYTICS**



**PRODUCTION
PLANNING**

DEFINITION

Collect inventory,
supplier, procurement
and production data



Calculate EOQ,
Safety Stock and
Reorder Point.



Evaluate supplier
performance through
lead time, quality,
and delivery metrics



Analyze spend,
procurement KPIs
and supplier risk.



Support production
scheduling and
capacity planning.

INCORPORATION TO RPSs' ENVIRONMENT

Material Master data
and Inventory records.



Inventory planning for
steel stock , tooling and
consumables



Supplier monitoring for
SKF, Sandvik , THK and
Tata Steel



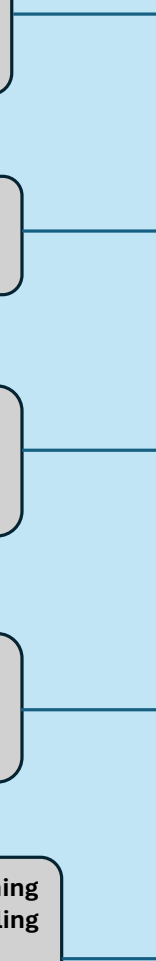
Spend Analysis.
Supplier Dashboard and
Sourcing Visibility



Manufacturing and planning
across CNC Turning , Milling
and Inspection

EXECUTIVE DASHBOARD

Enables real-time
monitoring of supply
chain efficiency,
procurement
performance,
inventory health, and
manufacturing
utilization .

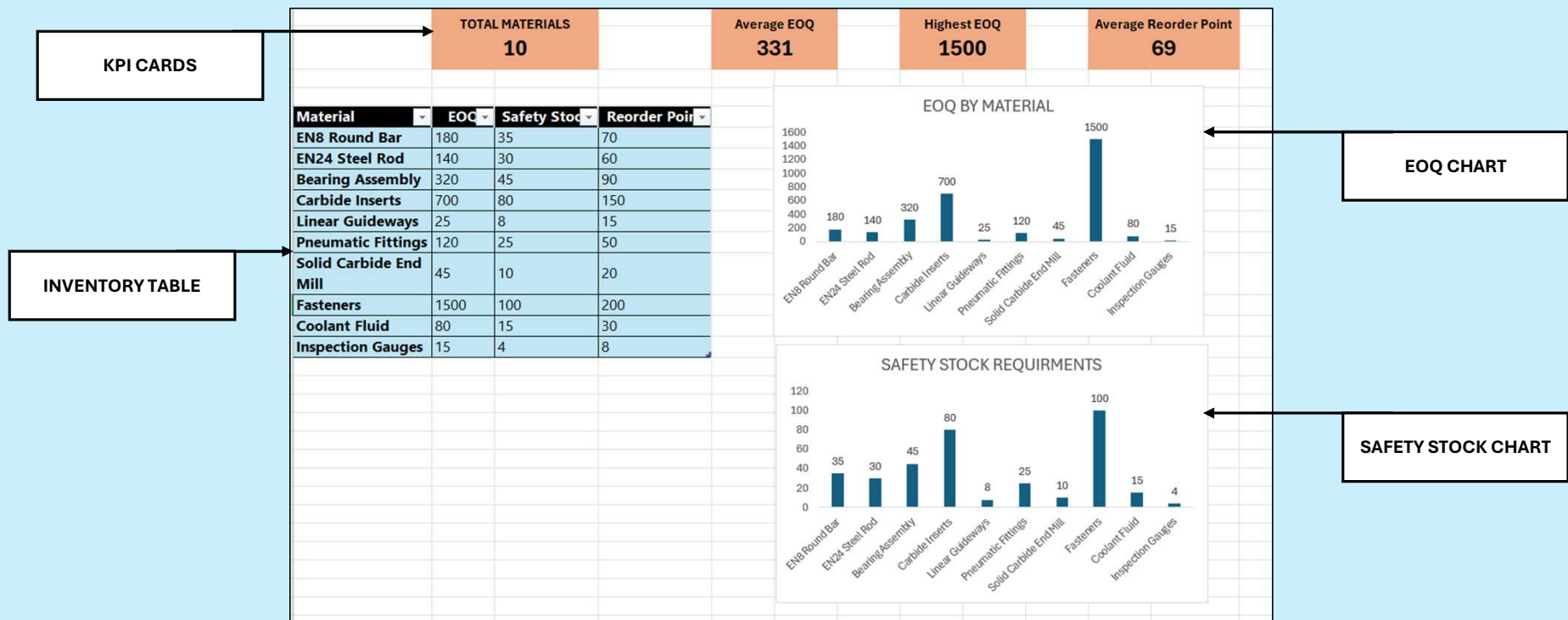


METHODOLOGY

The framework combines **inventory optimization, supplier analytics, procurement monitoring, and production planning** to transform fragmented operational data into actionable insights.

Using **EOQ analysis, safety stock calculations, supplier performance evaluation, and capacity planning** the system supports operational and strategic decision-making.

The results are consolidated into a centralized **Executive Dashboard** that enables real-time monitoring of inventory health, procurement efficiency, supplier reliability, and production performance.



SAMPLE of Executive Optimization Dashboard

RESULTS: Reflection on Design Objectives

- ✓ Reduced inventory carrying-cost exposure through optimized EOQ and safety-stock policies
- ✓ Improved supplier-performance visibility, helping minimize production disruptions and associated revenue losses
- ✓ Enabled procurement-spend monitoring and sourcing optimization opportunities
- ✓ Improved alignment between material availability and production schedules
- ✓ Created a decision-support framework capable of improving profitability through better operational and supply-chain decisions

DISCUSSIONS

The project demonstrates how manufacturing organizations can use integrated operational data to support inventory, procurement, supplier, and production decisions. Rather than optimizing individual functions independently, the framework adopts a systems approach where operational improvements across the value chain contribute to improved efficiency, reduced costs, and stronger business performance.